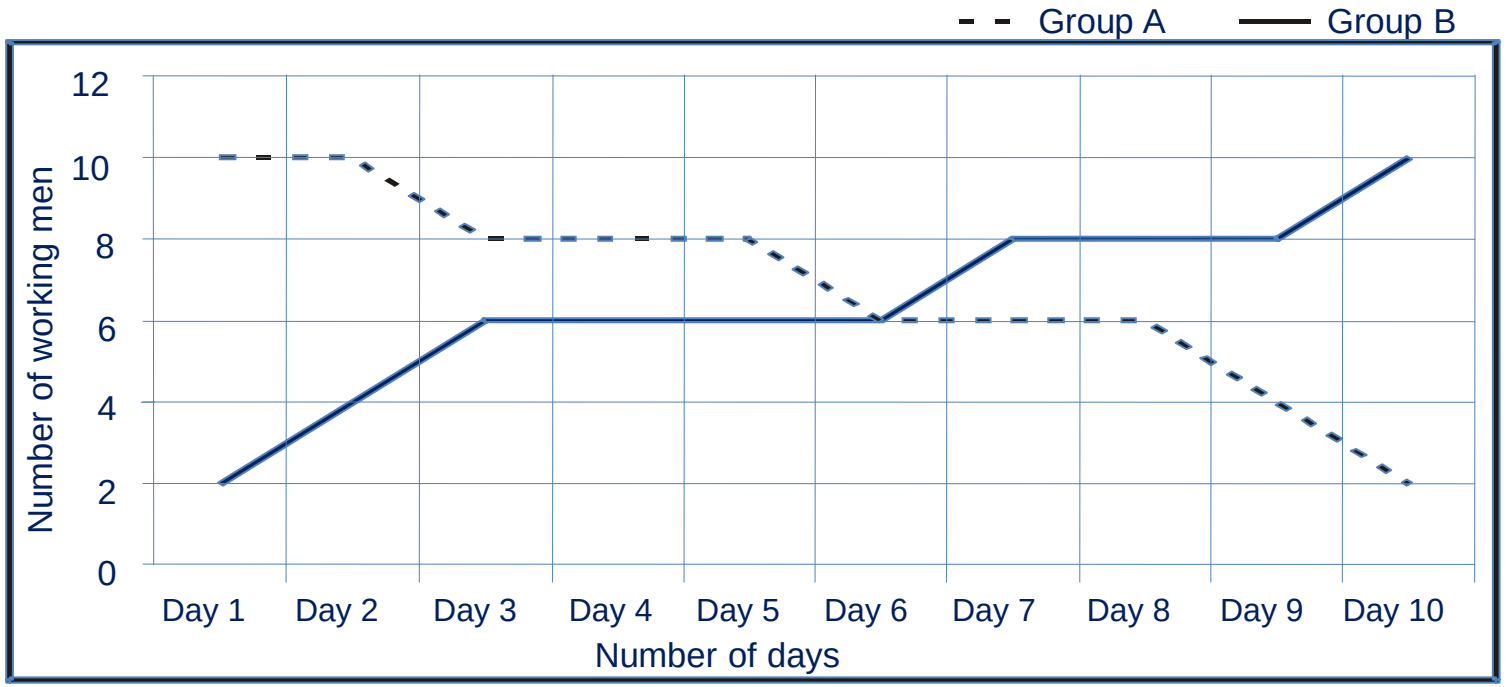
Career Launcher MBA

QA Algebra

Review Test – 01

Two groups A and B started working on two different projects and finished the projects in a 10 day timeline. The number of men vis-à-vis number of days they worked is shown in the graph below. Calculate the ratio of total amount of work done by the two respective groups using the graph plotted below. Assume that each person in both the groups is equally efficient as any other person in any group.



(1) 12 : 17

(2) 11 : 16

(3) 17 : 16

(4) 31 : 29

Question – 1

Solve for x such that $|x + 9| < |2x - 1| + |x + 3|$

(1) $x \in (-\infty, -3.5) \cup (2.5, \infty)$

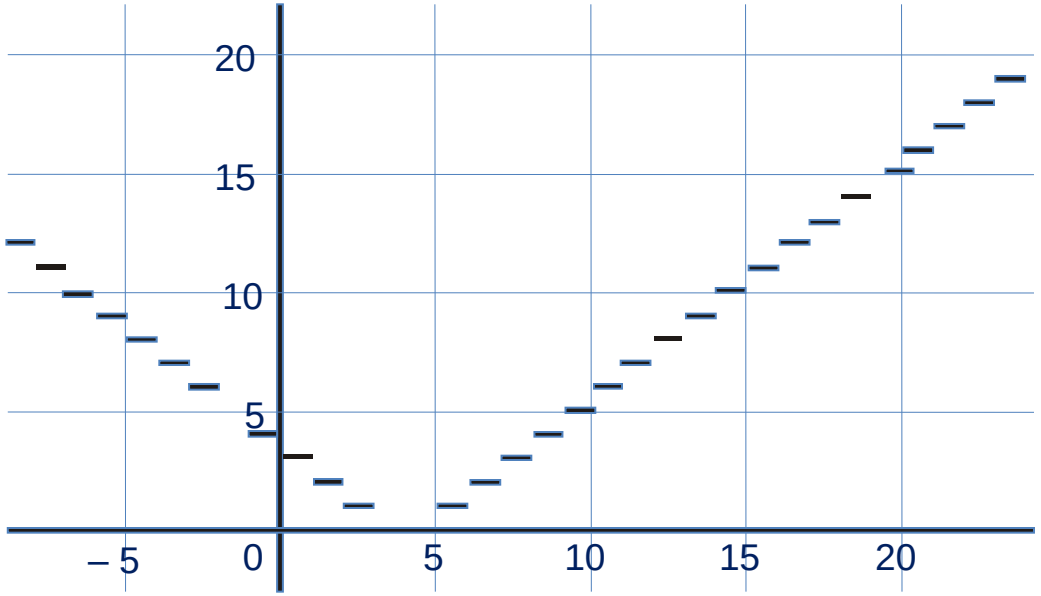
(2) $x \in (-\infty, -2.5) \cup (3.5, \infty)$

(3) $x \in (-\infty, 3.5)$

(4) $x \in (-\infty, -2.5)$

Question – 2

The graph given below is represented by which of the following equations



(1) f(x) = |[x – 5]|

(2) f(x) = |[x – 4]|

(3) f(x) = [|x – 5|]

(4) f(x) = [|x – 4|]

Where, [.] denotes the Greatest Integer Function/Floor Function and | . | denotes Modulus Function

Question – 3

$f(x) = \Sigma(-i)^{i+1} x^{n-i+1}$ defined for $0 < i < n + 2$, is a polynomial of degree n whose roots are ai where, $0 < i < n + 1$

Then find $\Sigma\left[\frac{1}{(a_i)}\right]$

(1) –1

(2) 1

(3) $\frac{(n)^{n+1}}{(n+1)^{n+2}}$

(4) $(-n)^{n+1}$

Question – 4

If [.] denotes the greatest integer function, then for how many integral values of x in the interval [1, 5] will the following equation satisfy?

$x^2 - [x^2] = (x - [x])^2$

(1) None

(2) 5

(3) 21

(4) Infinitely many

Question – 5

A rectangular sheet of dimension 2d × d has its four corners cut in the form of squares such that the open box, so formed from the remaining sheet, has the maximum volume. Find the maximum possible value of the volume.

(1) d^3

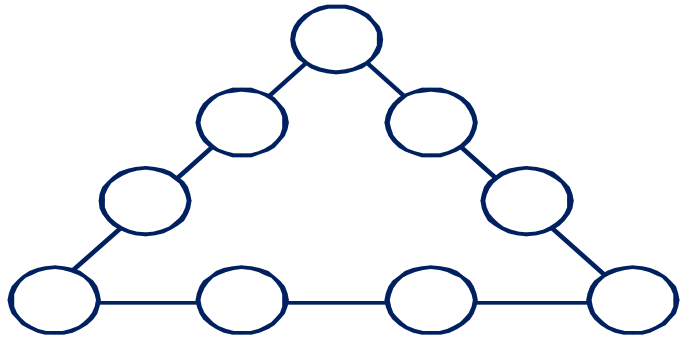
(2) $\frac{d^3}{2}$

(3) $\frac{d^3}{4}$

(4) $\frac{d^3}{16}$

Question – 6

In the given figure, each cell contains distinct digits from 1 to 9 such that the sum of digits along each side is same. Find the sum of the digits at the three vertices.



(1) 5

(2) 6

(3) 7

(4) 8

Question – 7

$f(x) = |x - 1| + |x - 2| + ... + |x - 50|$

For how many integral values of x, f(x) has minimum possible value and what is the minimum value of f(x)?

Question – 8

Find the area enclosed by graph of $\frac{|x|}{3}+\frac{|y|}{4}=1$

(1) 64

(2) 48

(3) 24

(4) 12

Question – 9

The minimum value of $(x + 3y + 5z + 7t) \times \left(\frac{1}{x} + \frac{1}{3y} + \frac{1}{5z} + \frac{1}{7t}\right)$ is

(1) 64

(2) 12

(3) 16

(4) 48

Question – 10

The maximum number of negative real roots and positive real roots of $f(x) = x^8 + 3x^5 - 8x^3 + 4x + 1$ are

(1) 3, 2

(2) 3, 4

(3) 4, 2

(4) 2, 1

Question – 11

Find the maximum volume of a right circular cylinder, if sum of radius and height is 15?

(1) 500π

(2) 250π

(3) 400π

(4) 200π

Question – 12

$3|K - 1| + K^2 - 7 > 0$

For what values of K will the given inequality be true?

(1) $(-\infty, -5) \cup (2, \infty)$

(2) $(-\infty, -1) \cup (2, \infty)$

(3) $(-\infty, -1) \cup (4, \infty)$

(4) $(-\infty, -5) \cup (4, \infty)$

Question – 13

$ax^4 + bx^3 + cx^2 + dx + e$ is divisible by $(x^2 - 3x + 2)$. Find the value of $17a + 9b + 5c + 3d + 2e$?

(1) 117

(2) 38

(3) 0

(4) 10

Question – 14

If $-7 < p < 5$ and $-4 < q < 6$, then find maximum possible value of $p^2 - pq + q^2$, where p and q are integers.

(1) 127

(2) 91

(3) 94

(4) 126

Question – 15

How many positive integer values are possible for x satisfying $(x - 6)(x - 8)(x - 10)...(x - 82) < 0$

(1) 39

(2) 38

(3) 24

(4) 25

Question – 16

How many natural numbers less than 60 can be expressed in the form of 2a + 3b, where a and b are natural numbers?

(1) 59

(2) 54

(3) 39

(4) None of these

Question – 17

p, q, r are integers such that $|p| < |q| < |r| < 40$ and $p + q + r = 20$ then what is the maximum value of pqr?

(1) 3610

(2) 3510

(3) 280

(4) 3500

Question – 18

How many real solutions exist for the equation $x^2 - 2^x = 0$

(1) 2

(2) 1

(3) 0

(4) More than 2

Question – 19

If $10^K = 12m + 10$, where m and k are integers, what is the value of $4^{(K-1)m}$?

(1) 64

(2) 4

(3) 1

(4) 16

Question – 20

If $m^{a \times b} < m^a \times m^b$, a, b are single digit natural numbers and m is a real number. Find the number of possible values of set {a, b}.

(1) 1

(2) 18

(3) 9

(4) 17

Question – 21

α and β are roots of the equation $x^2 - px + 12 = 0$. If the difference between the roots is at least 12. What are the possible value for p?

(1) $p > 8\sqrt{3}$

(2) $p < 8\sqrt{3}$

(3) $p \in (-\infty, -8\sqrt{3}] \cup [8\sqrt{3}, \infty)$

(4) $p \in (-\infty, -8\sqrt{3}) \cup (8\sqrt{3}, \infty)$

Question – 22

If a, b, c are distinct natural numbers less than 20 what is the maximum possible value of $|a - b| + |b - c| - |c - a|$?

(1) 20

(2) 40

(3) 35

(4) 34

Question – 23

In a list of 7 integers, one integer x is unknown. The other six integers are 20, 4, 10, 4, 8 and 4. If the mean, median and mode of these 7 integers are arranged in ascending order, they form an A.P. having all distinct numbers. The sum of all possible values of x is

(1) 40

(2) 34

(3) 38

(4) 32

Question – 24

If a > 0, then $2 \cdot \log_{a^2} 2 + 3 \cdot \log_{a^3} 3 + \ldots + n \cdot \log_{a^n} n = ?$

(1) $\log_a n$

(2) $\log_a \frac{n(n+1)}{2}$

(3) $\frac{n(n+1)(2n+1)}{6}$

(4) $\log_a n!$

Question – 25

Sum of n terms of the series:

$$\frac{3}{1^2.\,2^2}+\frac{5}{2^2.\,3^2}+\frac{7}{3^2.\,4^2}+\ldots$$

(1) $\frac{2n+n^2}{(n+1)^2}$

(2) n + 1

(3) $\frac{n^3+2n}{(n^2+1)^2}$

(4) $\frac{n^3+2n^2}{(n^2+1)^2}$

Question – 26

Find the solution set for [x] + [2x] + [3x] + [4x] = 14, where x is a real number and [x] represents the greatest integer less than or equal to x.

(1) $x < \frac{5}{3}$

(2) $\frac{3}{2} \le x < \frac{5}{3}$

(3) $1 \le x < \frac{4}{3}$

(4) None of these

Question – 27

An arithmetic series a_1, a_2, a_3,..., a_{1000} exists where no two consecutive terms are same. Find the value of $(a_{999}^2 – a_2^2)/(a_{501}^2 – a_{500}^2)$

(1) 1

(2) Depends on common difference

(3) 499

(4) 997

Question – 28

a, b, c are three consecutive integers between –5 and 5. How many different integral values can the equation $\frac{a^3+b^3+c^3+3abc}{(a+b+c)}$ take, if $b \neq 0$?

(1) 0

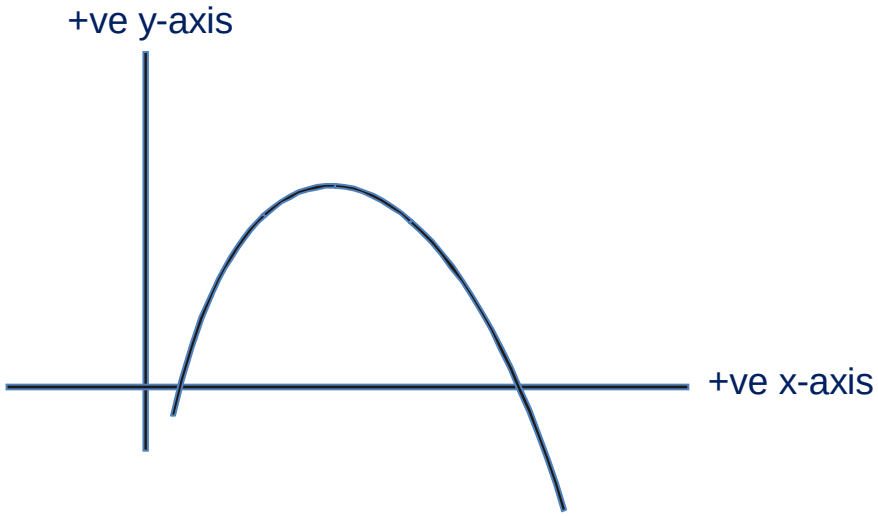
(2) 2

(3) 1

(4) 3

Question – 29

The following curve represents a quadratic function $y = ax^2 + bx + c$. Which of the following cannot be the value of c if a = –5 (Figure drawn on scale)



(1) 2 (2) –2 (3) –10 (4) –5

Question – 30

Thanks!